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ME5180: Dynamics
Project 01 Calculations
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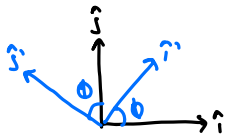
Inertial Frame: x, y, z
Rotating Frame: x', y', z' $\Big\}$ $z' = z$

At time t rotating frame rotates by $\phi(t) = \Omega t$ about z-axis relative to inertial frame

Say $\vec{v} = x'\hat{i}' + y'\hat{j}' = x\hat{i} + y\hat{j}$



$$\hat{i}' = \cos\phi\,\hat{i} + \sin\phi\,\hat{j}$$
$$\hat{j}' = \cos(90^\circ + \phi)\hat{i} + \sin(90^\circ + \phi)\hat{j} = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}$$

So if we have:

$$\vec{v} = x'\hat{i}' + y'\hat{j}'$$
$$= x'(\cos\phi\,\hat{i} + \sin\phi\,\hat{j}) + y'(-\sin\phi\,\hat{i} + \cos\phi\,\hat{j})$$
$$= x'\cos\phi\,\hat{i} + x'\sin\phi\,\hat{j} - y'\sin\phi\,\hat{i} + y'\cos\phi\,\hat{j}$$
$$= (x'\cos\phi - y'\sin\phi)\hat{i} + (x'\sin\phi + y'\cos\phi)\hat{j}$$

So original coordinates:

$$x = x'\cos\phi - y'\sin\phi$$
$$y = x'\sin\phi + y'\cos\phi$$
$$z = z'$$

Therefore:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_z(\phi(t)) = R_z(\Omega t)} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}$$

* Rotation matrix for converting positions between rotating and inertial reference frames.

Pivot Position (in the Rotating Frame)

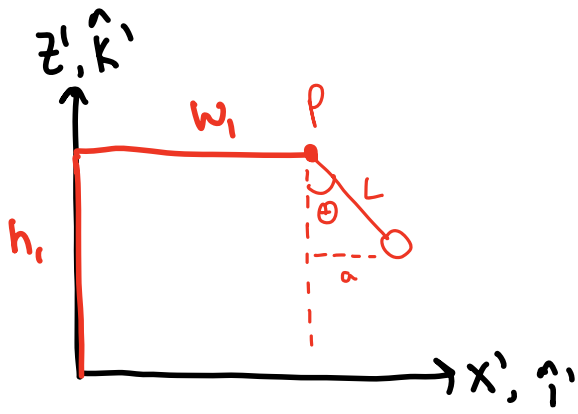
$$\vec{r}'_{Pivot} = \begin{bmatrix} w_1 \\ 0 \\ h_1 \end{bmatrix}$$

Pivot Position (in the Inertial Frame)

$$\vec{r}_{Pivot}(t) = R_z(\phi)\,\vec{r}'_{Pivot}$$

$$= \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ 0 \\ h_1 \end{bmatrix} = \begin{bmatrix} w_1 \cos\phi \\ w_1 \sin\phi \\ h_1 \end{bmatrix} = \begin{bmatrix} w_1 \cos(\Omega t) \\ w_1 \sin(\Omega t) \\ h_1 \end{bmatrix}$$

Bob Position (in the Rotating Frame) *Relative to Pivot, P*



$$\vec{r}'_{Bob,rel} = \begin{bmatrix} L\sin\theta \\ 0 \\ -L\cos\theta \end{bmatrix}$$

Bob Position (in the Inertial Frame)

$$\vec{r}_{Bob,rel}(t) = R_z(\phi)\,\vec{r}'_{Bob,rel}$$

$$= \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} L\sin\theta \\ 0 \\ -L\cos\theta \end{bmatrix} = \begin{bmatrix} L\sin\theta\cos\phi \\ L\sin\theta\sin\phi \\ -L\cos\theta \end{bmatrix} = \begin{bmatrix} L\sin\theta\cos(\Omega t) \\ L\sin\theta\sin(\Omega t) \\ -L\cos\theta \end{bmatrix}$$

Bob Position (absolute) in the inertial frame

$$\vec{r}_{Bob}(t) = \vec{r}_{Pivot}(t) + \vec{r}_{Bob,rel}(t)$$

$$= \begin{bmatrix} W_1 \cos(\Omega t) \\ W_1 \sin(\Omega t) \\ h_1 \end{bmatrix} + \begin{bmatrix} L \sin\theta \cos(\Omega t) \\ L \sin\theta \sin(\Omega t) \\ -L\cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} W_1 \cos(\Omega t) + L\sin\theta\cos(\Omega t) \\ W_1 \sin(\Omega t) + L\sin\theta \sin(\Omega t) \\ h_1 - L\cos\theta \end{bmatrix} = \begin{bmatrix} \cos(\Omega t)\left[W_1 + L\sin\theta(t)\right] \\ \sin(\Omega t)\left[W_1 + L\sin\theta(t)\right] \\ h_1 - L\cos\theta(t) \end{bmatrix}$$

$$\boxed{\begin{aligned} x(t) &= \cos(\Omega t)\left[W_1 + L\sin\theta(t)\right] \\ y(t) &= \sin(\Omega t)\left[W_1 + L\sin\theta(t)\right] \\ z(t) &= h_1 - L\cos\theta(t) \end{aligned}}$$

Let's say:

$$R(t) = w_1 + L\sin\theta(t)$$

So:

$$x(t) = R\cos(\Omega t)$$
$$y(t) = R\sin(\Omega t)$$
$$z(t) = h_1 - L\cos\theta(t)$$

r_dot = Differential(t)(R) or expand_derivatives(differentiate(R, t))

$$\boxed{\dot{R}(t) = L\cos\theta(t)\,\dot{\theta}(t)}$$

x_dot = expand_derivatives(differentiate(x, t))

$$\dot{x}(t) = \dot{R}\cos(\Omega t) - R\Omega\sin(\Omega t)$$

$$\boxed{\dot{x}(t) = L\cos\theta(t)\dot{\theta}(t)\cos(\Omega t) - \Omega(w_1 + L\sin\theta(t))\sin(\Omega t)}$$

$$\dot{y}(t) = \dot{R}\sin(\Omega t) + R\Omega\cos(\Omega t)$$

y_dot = expand_derivatives(differentiate(y, t))

$$\boxed{\dot{y}(t) = L\cos\theta(t)\dot{\theta}(t)\sin(\Omega t) + \Omega(w_1 + L\sin\theta(t))\cos(\Omega t)}$$

$$\boxed{\dot{z}(t) = L\sin\theta(t)\,\dot{\theta}(t)}$$

z_dot = expand_derivatives(differentiate(z, t))

$$v^2 = \underbrace{\dot{x}^2 + \dot{y}^2} + \dot{z}^2$$

$$\left(\dot{R}\cos(\Omega t) - R\Omega\sin(\Omega t)\right)^2 + \left(\dot{R}\sin(\Omega t) + R\Omega\cos(\Omega t)\right)^2$$

$$= \dot{R}^2\cos^2(\Omega t) - 2\dot{R}R\Omega\cos(\Omega t)\sin(\Omega t) + \Omega^2R^2\sin^2(\Omega t)$$
$$+$$
$$\dot{R}^2\sin^2(\Omega t) + 2\dot{R}R\Omega\sin(\Omega t)\cos(\Omega t) + R^2\Omega^2\cos^2(\Omega t)$$

$$= \dot{R}^2\cos^2(\Omega t) + \Omega^2R^2\sin^2(\Omega t) + \dot{R}^2\sin^2(\Omega t) + R^2\Omega^2\cos^2(\Omega t)$$

$$= \dot{R}^2\left(\cos^2(\Omega t) + \sin^2(\Omega t)\right) + R^2\Omega^2\left(\cos^2(\Omega t) + \sin^2(\Omega t)\right)$$

$$\dot{x}^2 + \dot{y}^2 = \dot{R}^2 + R^2\Omega^2$$
$$= L^2\cos^2\theta(t)\,\dot{\theta}(t)^2 + \Omega^2(w_1 + L\sin\theta(t))^2$$

$$\dot{z}^2 = L^2\sin^2\theta(t)\,\dot{\theta}(t)^2$$

$$v^2 = L^2\cos^2\theta(t)\,\dot{\theta}(t)^2 + \Omega^2(w_1 + L\sin\theta(t))^2 + L^2\sin^2\theta(t)\,\dot{\theta}(t)^2$$

$$= L^2\dot{\theta}(t)^2\left[\cos^2\theta(t) + \sin^2\theta(t)\right] + \Omega^2(w_1 + L\sin\theta(t))^2$$

$$\boxed{v^2 = L^2\dot{\theta}(t)^2 + \Omega^2(w_1 + L\sin\theta(t))^2}$$

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V2 = simplify(xdot^2 + ydot^2 + zdot^2)
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$$T = \frac{1}{2}mv^2$$

$$\boxed{T = \frac{1}{2}m\left[L^2\dot{\theta}(t)^2 + \Omega^2(w_1 + L\sin\theta(t))^2\right]}$$

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T = 0.5*m*V2 |> expand or simplify
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$$V = mgz$$

$$\boxed{V = mg(h_1 - L\cos\theta(t))}$$

$$L(\theta,\dot{\theta}) = T - V \quad \leftarrow \text{logr} = T - V$$

$$= \frac{1}{2}m\left[L^2\dot{\theta}(t)^2 + \Omega^2(w_1 + L\sin\theta(t))^2\right] - mg(h_1 - L\cos\theta(t))$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{\partial L}{\partial \dot{\theta}} = mL^2\dot{\theta}(t) \quad \leftarrow \text{dL_dThetadot} = \text{differentiate}(\text{Lagr}, \text{Thetadot})$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = mL^2\ddot{\theta}(t)$$

Say $f(\theta) = w_1 + L\sin\theta$:

$$\frac{1}{2}m\Omega^2(w_1 + L\sin\theta)^2 = \frac{1}{2}m\Omega^2 f(\theta)^2$$

$$\frac{\partial}{\partial\theta}\left[\frac{1}{2}m\Omega^2 f(\theta)^2\right] = m\Omega^2 f(\theta) f'(\theta)$$

$$\hookrightarrow f'(\theta) = L\cos\theta$$

$$\frac{\partial}{\partial\theta}\left[\frac{1}{2}m\Omega^2(w_1 + L\sin\theta)^2\right] = m\Omega^2(w_1 + L\sin\theta)(L\cos\theta)$$

now, $\frac{\partial}{\partial\theta}\left[-mg(h_1 - L\cos\theta)\right] = -mgL\sin\theta$

$$\frac{\partial L}{\partial \theta} = m\Omega^2(w_1 + L\sin\theta)(L\cos\theta) - mgL\sin\theta \quad \leftarrow \text{dL_dTheta} = \text{differentiate}(\text{Lagr}, \theta)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

$$\hookrightarrow mL^2\ddot{\theta}(t) - m\Omega^2(w_1 + L\sin\theta)(L\cos\theta) + mgL\sin\theta = 0$$

$$L\ddot{\theta}(t) - \Omega^2(w_1 + L\sin\theta)\cos\theta + g\sin\theta = 0$$

$$L\ddot{\theta}(t) + g\sin\theta(t) - \Omega^2(w_1 + L\sin\theta(t))\cos\theta(t) = 0$$

$$\boxed{\ddot{\theta}(t) = \frac{\Omega^2}{L}(w_1 + L\sin\theta(t))\cos\theta(t) - \frac{g}{L}\sin\theta(t)}$$

$$\text{eq} = \text{ddt_dL_dThetadot} - \text{dL_dTheta} \sim 0$$

or

$$\text{Solve_for}(\text{eq}, \text{Thetaddot})$$

$$\omega_0 = \sqrt{\frac{g}{L}} = \sqrt{\frac{9.81}{0.15}} \approx 8.08 \text{ rad/s}$$

↑ Natural Frequency / Swing Rate

Let's say $\Omega_{slow} = 1$ rad/s

$\Omega_{fast} = 12$ rad/s

Start at rest, $\theta_0 = 0$? or Small displacement $\theta_0 = 0.05$ rad ⇝ Both?

<u>Computational Trick</u>

$$\ddot{\theta} = f(\theta, \dot{\theta})$$

Say: $u_1 = \theta$, $u_2 = \dot{\theta}$ $\Rightarrow$ $\dot{u}_1 = \dot{\theta} = u_2$, $\dot{u}_2 = \ddot{\theta} = f(u_1, u_2)$

* Must decompose 2nd order ODE into 1st order system for Julia to solve.

$$\dot{u}_1 = u_2$$

$$\dot{u}_2 = \frac{\Omega^2}{L}(w_1 + L\sin u_1)\cos u_1 - \frac{g}{L}\sin u_1$$